

Name: _____



Exit quiz

Converting fractions to recurring decimals

1 A decimal with an infinite repeating pattern of digits is called a _____ decimal. Fill in the blank

2 Match each decimal given using dot notation to its equivalent decimal. Write the correct letter in each box

a	$0.\dot{1}\dot{2}$
b	$0.12\dot{3}$
c	$0.\dot{1}2\dot{3}$
d	$0.\dot{1}$
e	$0.1\dot{2}\dot{3}$

	0.123333333...
	0.111111111...
	0.12121212...
	0.123123123...
	0.123232323...

3 Select all the fractions which are equivalent to a recurring decimal. Tick **3** correct answers

☐ $\frac{1}{3}$

☐ $\frac{1}{12}$

☐ $\frac{3}{24}$

☐ $\frac{6}{15}$

☐ $\frac{10}{12}$



4 Use short division to write $\frac{8}{15}$ as a decimal. Tick **1** correct answer

☐ 0.53

☐ 0.5333333333

☐ 0.5 $\dot{3}$

☐ 0. $\dot{5}\dot{3}$

5 Jacob uses his calculator to write $\frac{11}{12}$ as a decimal. His calculator display shows the number 0.9166666667. Jacob says, "This shows my fraction terminates." Is Jacob correct? Explain why. Tick **1** correct answer

☐ Yes; Jacob's number has 10 digits after the decimal point so it terminates.

☐ Yes; the last digit is 7 not 6 so it doesn't repeat. So it must terminate.

☐ No; the calculator only has a 12 digit display, the last 6 is rounded up to 7.

☐ No; all decimals recur eventually.

6 Izzy writes the recurring decimal 0.71 $\dot{6}$ as the fraction $\frac{\square}{60}$. What number should she write in the square? _____ Fill in the blank

